



Chapter 5

Jacobians: Velocities and Static Forces

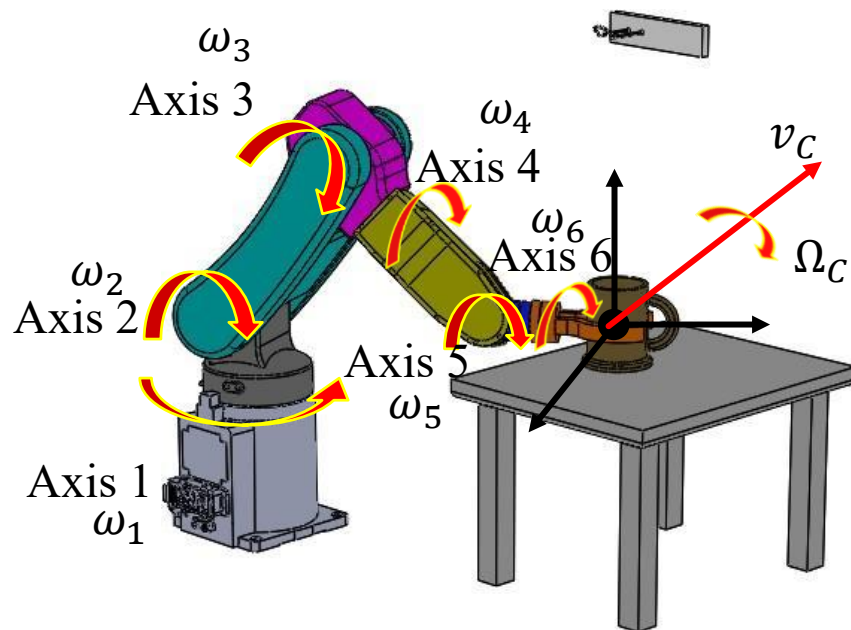
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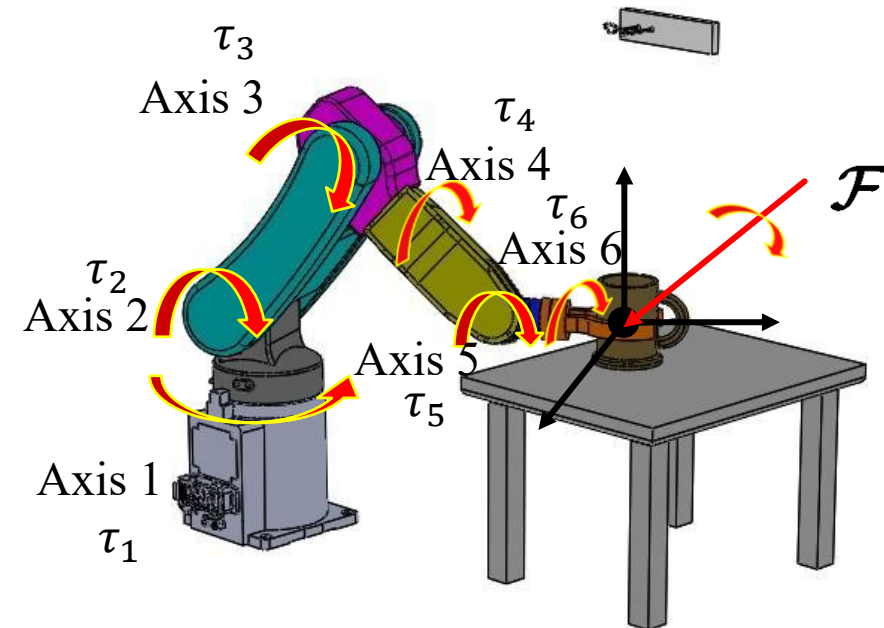
Questions



- During the motion of the robotic arm, how to determine the relationship between the **angular velocities of the joints** and the **velocity of end effector**?
- When there is an **external force on the end effector** of the robotic arm, how to determine the **torque of each joint**?



Determine the velocity of joints



Determine the torque of joints



Chapter 5 Jacobians Velocities and Static Forces

- 1. Time-varying Position and Orientation**
- 2. Linear and Rotational Velocity of Rigid Bodies**
- 3. Velocity “Propagation” from Link to Link**
- 4. Jacobians**
- 5. Static Forces in Manipulators**
- 6. Jacobian in the Force Domain**
- 7. Cartesian Transformation of Velocities and Static Forces**



1. Time-varying Position and Orientation

□ Differentiation of a position vector P_Q

Derivative of position vector ${}^B P_Q$ in the frame $\{B\}$

$${}^B V_Q = \frac{d}{dt} {}^B P_Q = \lim_{\Delta t \rightarrow 0} \frac{{}^B P_Q(t + \Delta t) - {}^B P_Q(t)}{\Delta t}$$

When expressed in frame $\{A\}$

$$\begin{aligned} {}^A ({}^B V_Q) &= {}^A \left(\frac{d}{dt} {}^B P_Q \right) \\ &= {}^A R^B ({}^B V_Q) = {}^A R^B \underline{{}^B V_Q} \end{aligned}$$

When both superscripts are the same

If Q is not changing in time relative to $\{B\}$, then the velocity calculated is zero - even if there is some other frame in which Q is varying.

A velocity vector is associated with a point in space, but the numerical values describing the velocity of that point depend on two frames: one with respect to which the differentiation was done, and one in which the resulting velocity vector is expressed

Very often consider the velocity of the *origin of a frame* relative to some universe reference frame, and for this special case, define a shorthand notation

$v_C = {}^U V_{C \text{ Org}}$ is the velocity of the origin of frame $\{C\}$ relative to the universe reference frame $\{U\}$



1. Time-varying Position and Orientation

□ **Ex:** Figure shows a fixed universe frame $\{U\}$, a frame attached to a train traveling at 100 mph $\{T\}$, and a frame attached to a car traveling at 30 mph $\{C\}$. Both vehicles are heading in the X direction of $\{U\}$.

$${}^U V_T = 100\hat{i}$$

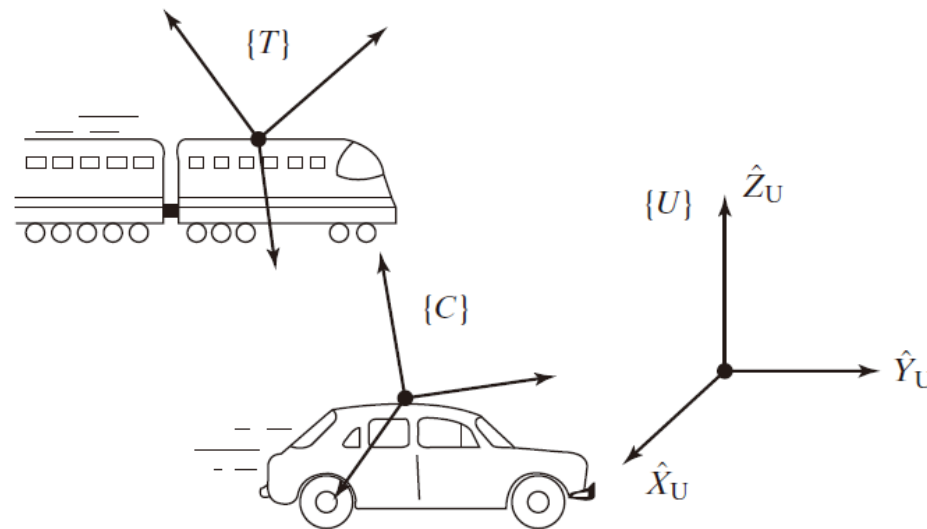
$${}^U V_C = 30\hat{i}$$

It has:

$${}^U \left(\frac{d}{dt} {}^U P_{C\text{ ORG}} \right) = {}^U V_{C\text{ ORG}} = v_C = 30\hat{i}$$

$${}^C ({}^U V_{T\text{ ORG}}) = {}^C v_T = {}^C R(v_T) = {}^C R(100\hat{i}) = {}^U R^{-1} 100\hat{i}$$

$$\begin{aligned} {}^C ({}^T V_{C\text{ ORG}}) &= {}^C R({}^T V_{C\text{ ORG}}) \\ &= {}^C R^U R(-70\hat{i}) = -{}^U R^{-1} {}^U R 70\hat{i} \end{aligned}$$



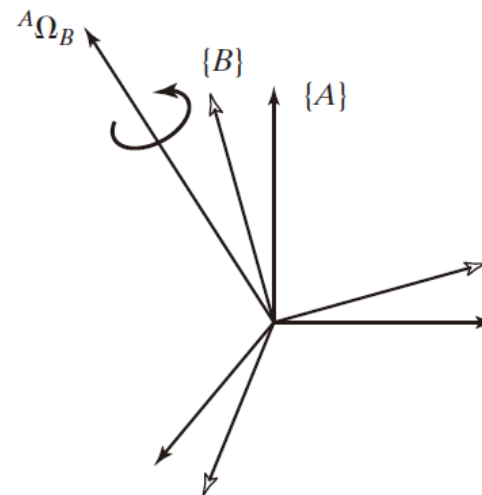
Example of some frames in linear motion



1. Time-varying Position and Orientation

□ Angular velocity vector ${}^A\Omega_B$

- The rotation of frame $\{B\}$ relative to frame $\{A\}$
- Direction of ${}^A\Omega_B$: the instantaneous **axis** of rotation
- Magnitude of ${}^A\Omega_B$: the **speed** of rotation



Frame $\{B\}$ is rotating with angular velocity ${}^A\Omega_B$ relative to frame $\{A\}$

An angular velocity vector may be expressed in any coordinate system: e.g., ${}^C({}^A\Omega_B)$ is the angular velocity of frame $\{B\}$ relative to $\{A\}$ expressed in terms of frame $\{C\}$.

Simplified notation of angular velocity of frame $\{C\}$ relative to the universe reference frame $\{U\}$

$$\omega_C = {}^U\Omega_C$$

${}^A\omega_C$ is the angular velocity of frame $\{C\}$ expressed in terms of $\{A\}$ (though the angular velocity is with respect to $\{U\}$)



2. Linear and Rotational Velocity of Rigid Bodies

- Investigate the description of **motion of a rigid body** as far as **velocity** is concerned: the **motion of rigid bodies** can be equivalently studied as the **motion of frames relative to one another**
- **Linear Velocity:** A frame $\{B\}$ attached to a rigid body, describe the motion of ${}^B Q$ relative to a fixed frame $\{A\}$

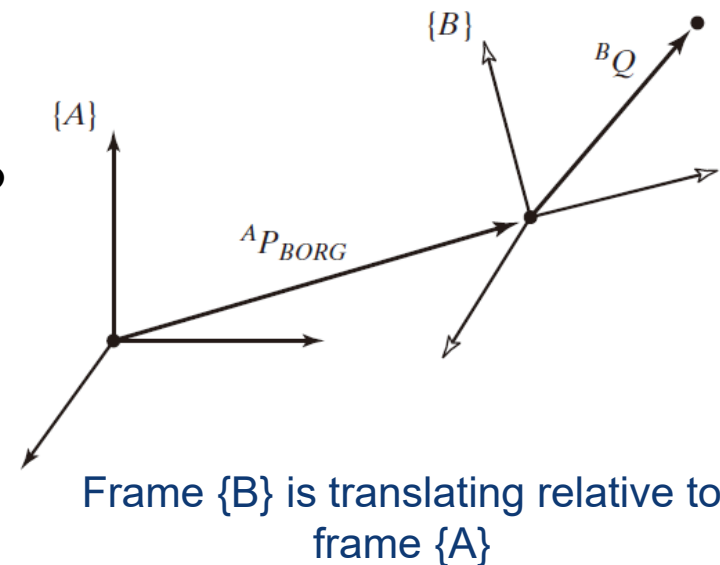
${}^A P_{BORG}$: position vector

${}^A_B R$: rotation matrix

Linear movement: ${}^A_B R$ not changing with time, motion of point Q relative to $\{A\}$ is due to ${}^A P_{BORG}$ or ${}^B Q$ changing in time.

Linear velocity of point Q :

$${}^A V_Q = {}^A V_{BORG} + {}^A_B R {}^B V_Q$$



Linear movement: Relative orientation of $\{B\}$ and $\{A\}$ remains constant but ${}^A P_{BORG}$ can change

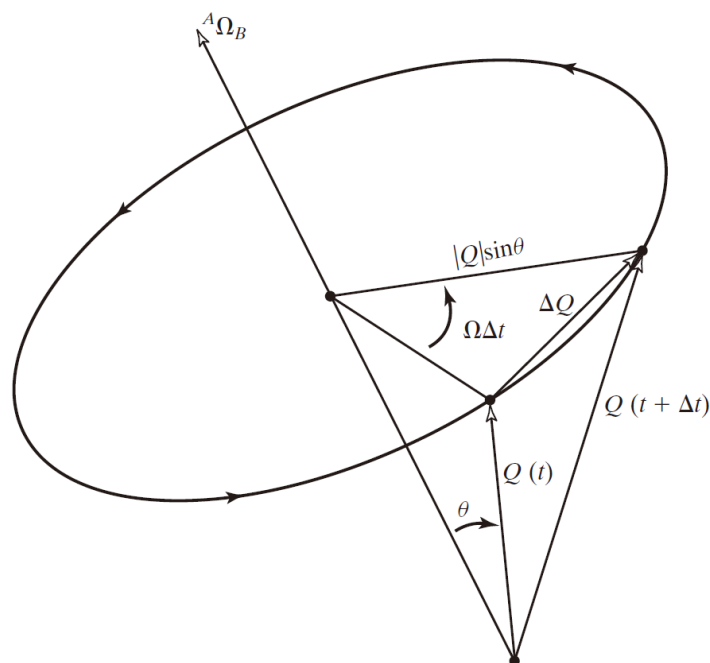


2. Linear and Rotational Velocity of Rigid Bodies

□ **Rotational Velocity:** two frames with **coincident origins** and with **zero linear relative velocity**

The orientation of frame $\{B\}$ with respect to frame $\{A\}$ is changing in time.

Rotational velocity of $\{B\}$ relative to $\{A\}$ is described by a vector ${}^A\Omega_B$ (with **orientation** and **magnitude**)



The velocity of a point due to an angular velocity

Case 1: consider that the vector Q is constant as viewed from frame $\{B\}$

$${}^B V_Q = 0$$

Point Q will have a velocity as seen from $\{A\}$ that is due to the rotational velocity ${}^A\Omega_B$

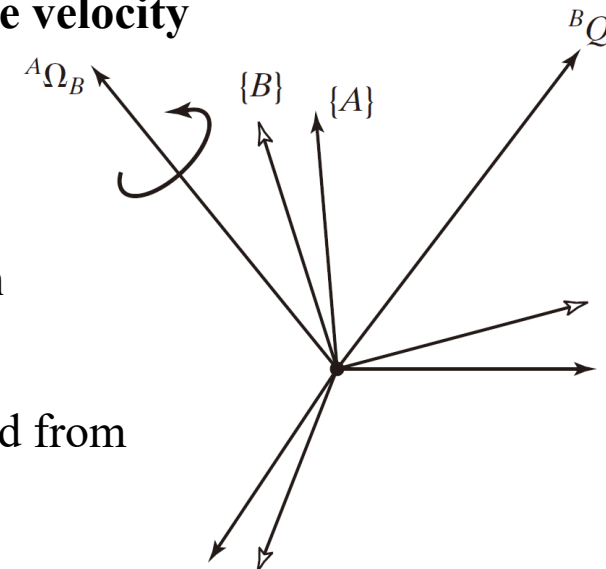
The magnitude of the differential change is

$$|\Delta Q| = (|{}^A Q| \sin \theta) (|{}^A \Omega_B| \Delta t)$$

The differential change in ${}^A Q$ must be perpendicular to both ${}^A \Omega_B$ and ${}^A Q$, and the orientation is

$${}^A V_Q = {}^A \Omega_B \times {}^A Q$$

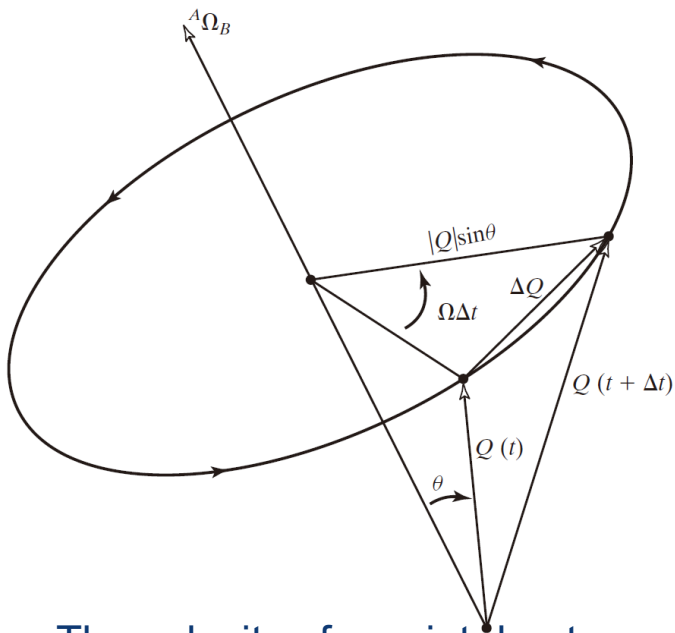
Note that: both ${}^A \Omega_B$ and ${}^A Q$ are vectors





2. Linear and Rotational Velocity of Rigid Bodies

□ Rotational Velocity (Continue)



The velocity of a point due to an angular velocity

Case 2: more general case and the vector Q could also be changing with respect to frame $\{B\}$, so, adding this component

$${}^A V_Q = \underline{{}^A ({}^B V_Q)} + {}^A \Omega_B \times {}^A Q$$

Direction due to the translation of Q in $\{A\}$

Using a rotation matrix to remove the dual-superscript

$${}^A V_Q = {}^A_B R {}^B V_Q + {}^A \Omega_B \times {}^A_B R {}^B Q$$

□ **Simultaneous Linear and Rotational Velocity:** adding the linear velocity of the origin ${}^A V_{BORG}$ to the rotational velocity

$${}^A V_Q = {}^A V_{BORG} + {}^A_B R {}^B V_Q + {}^A \Omega_B \times {}^A_B R {}^B Q$$

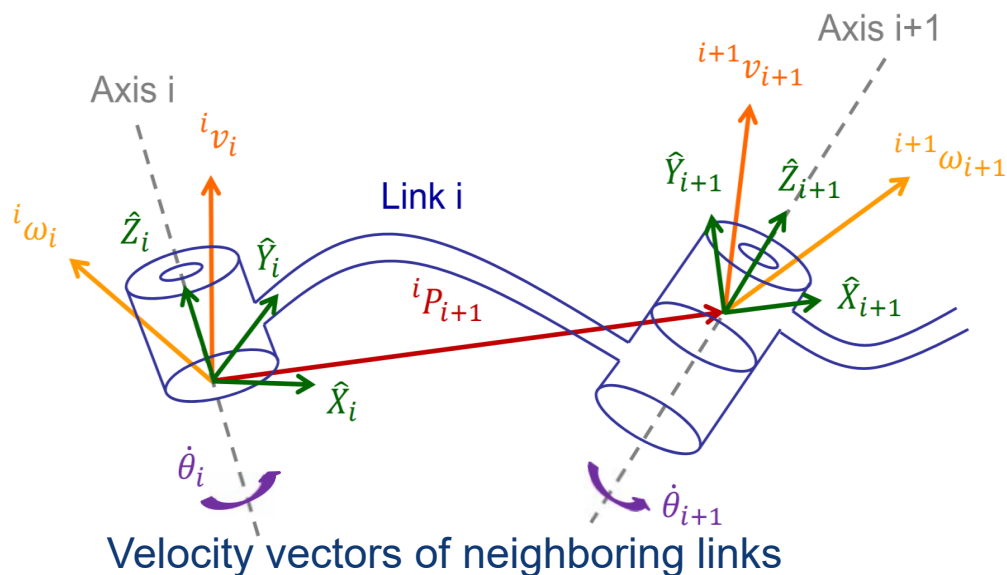
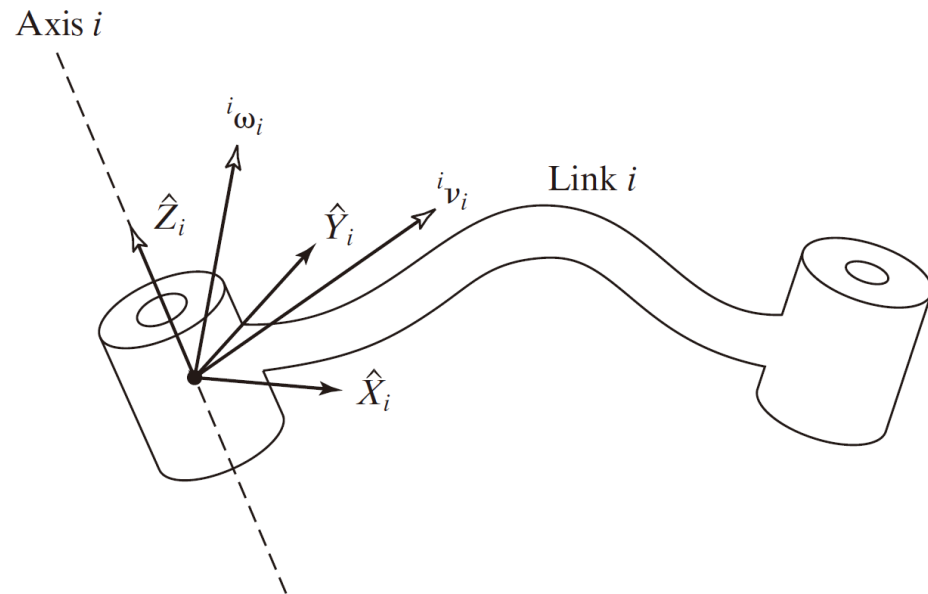
The general formula for velocity of a vector fixed in frame $\{B\}$, as seen from frame $\{A\}$

3. Velocity “Propagation” from Link to Link

□ In considering the motions of robot links, use link frame $\{0\}$ as reference frame

- v_i is the linear velocity of the origin of link frame $\{i\}$
- ω_i is the angular velocity of link frame $\{i\}$

At any instant, each link of a robot in motion has some **linear** and **angular** velocity.



- A manipulator is a chain of bodies, each one capable of motion relative to its neighbors. The velocity of link $i + 1$ will be that of link i , plus whatever new velocity components were added by joint $i + 1$.
- **From link to link:** take each link as a rigid body with linear and angular velocity vectors describing its motion. Express these velocities with respect to the link frame itself, rather than with respect to the base coordinate system.



3. Velocity “Propagation” from Link to Link

□ The angular velocity of link $i + 1$ is the same as that of link i , **plus a new component** caused by rotational velocity at joint $i + 1$.

□ Rotational joint written in terms of frame $\{i\}$

- Angular velocity propagation

$${}^i\omega_{i+1} = {}^i\omega_i + {}^{i+1}R\dot{\theta}_{i+1}{}^{i+1}\hat{Z}_{i+1}$$

Pre-multiplying
both sides

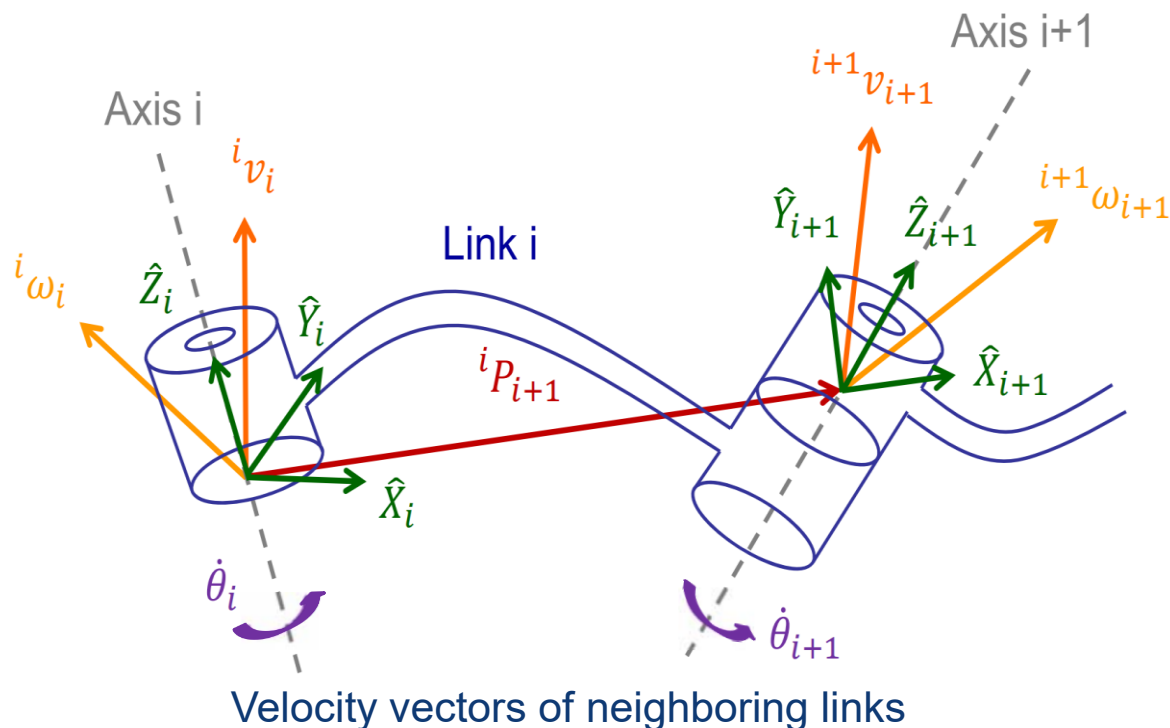
$${}^{i+1}\omega_{i+1} = {}^{i+1}R{}^i\omega_i + \dot{\theta}_{i+1}{}^{i+1}\hat{Z}_{i+1}$$

- Linear velocity propagation

$${}^i v_{i+1} = {}^i v_i + {}^i\omega_i \times {}^i P_{i+1}$$

Pre-multiplying
both sides

$${}^{i+1}v_{i+1} = {}^{i+1}R({}^i v_i + {}^i\omega_i \times {}^i P_{i+1})$$



Simultaneous Linear and Rotational Velocity

$${}^A V_Q = {}^A V_{BORG} + {}^A R {}^B V_Q + {}^A \Omega_B \times {}^A R {}^B Q$$

${}^A V_{BORG}$ is vanished because ${}^i P_{i+1}$ is constant in frame $\{i\}$.



3. Velocity “Propagation” from Link to Link

□ Prismatic joint

- Angular velocity propagation

$${}^i\omega_{i+1} = {}^i\omega_i$$

Pre-multiplying
both sides

$$\downarrow {}^{i+1}_i R$$

$${}^{i+1}\omega_{i+1} = {}^{i+1}_i R {}^i\omega_i$$

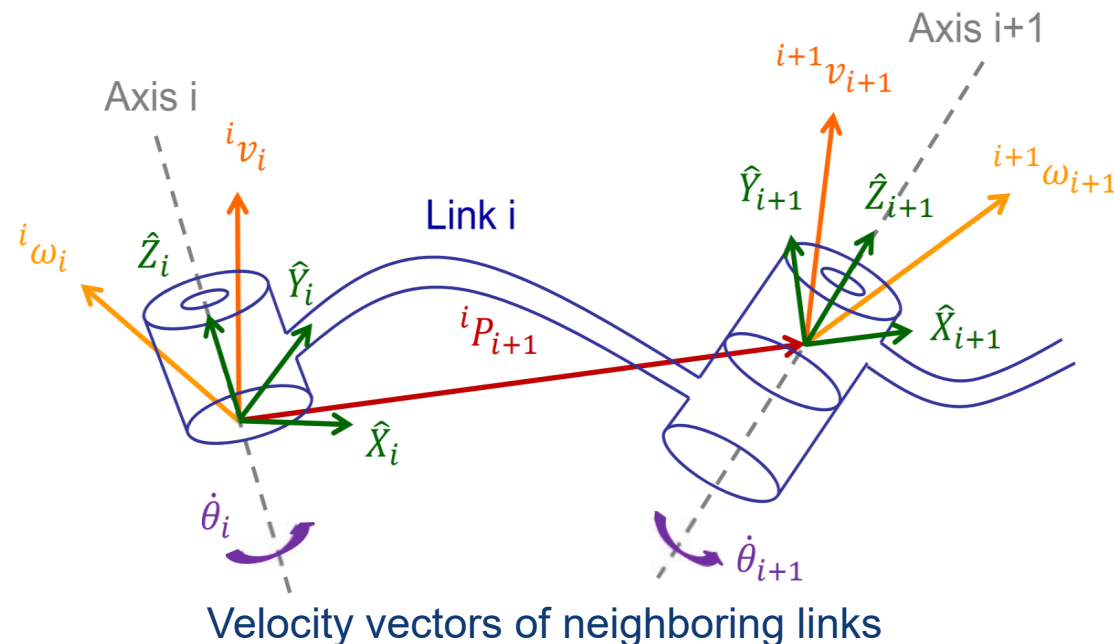
- Linear velocity propagation

$${}^i v_{i+1} = ({}^i v_i + {}^i \omega_i \times {}^i P_{i+1}) + \frac{{}^i R \dot{d}_{i+1} {}^{i+1} \hat{Z}_{i+1}}{\dot{d}_{i+1} {}^{i+1} \hat{Z}_{i+1}}$$

Pre-multiplying
both sides

$$\downarrow {}^{i+1}_i R$$

$${}^{i+1} v_{i+1} = {}^{i+1}_i R ({}^i v_i + {}^i \omega_i \times {}^i P_{i+1}) + \dot{d}_{i+1} {}^{i+1} \hat{Z}_{i+1}$$



- Applying these equations successively from link to link, can compute the rotational and linear velocities of the last link ${}^N\omega_N$ and ${}^N v_N$.
- If the velocities are desired in terms of the **base coordinate system**, they can be rotated into base coordinates by **multiplication with ${}^0_N R$** .



4. Jacobians

- The Jacobian is a **multidimensional form of the derivative**, e.g., that we have six functions, each of which is a function of six independent variables

$$y_1 = f_1(x_1, x_2, x_3, x_4, x_5, x_6)$$

$$y_2 = f_2(x_1, x_2, x_3, x_4, x_5, x_6)$$

⋮

$$y_6 = f_6(x_1, x_2, x_3, x_4, x_5, x_6)$$

Vector notation $\Rightarrow Y = F(X)$

- Calculating the differentials of y_i as a function of differentials of x_j , using chain rule

$$\delta y_1 = \frac{\partial f_1}{\partial x_1} \delta x_1 + \frac{\partial f_1}{\partial x_2} \delta x_2 + \dots + \frac{\partial f_1}{\partial x_6} \delta x_6$$

$$\delta y_2 = \frac{\partial f_2}{\partial x_1} \delta x_1 + \frac{\partial f_2}{\partial x_2} \delta x_2 + \dots + \frac{\partial f_2}{\partial x_6} \delta x_6$$

⋮

$$\delta y_6 = \frac{\partial f_6}{\partial x_1} \delta x_1 + \frac{\partial f_6}{\partial x_2} \delta x_2 + \dots + \frac{\partial f_6}{\partial x_6} \delta x_6$$

$\Rightarrow \delta Y = \frac{\partial F}{\partial X} \delta X = \underline{J(X)} \delta X$ Jacobian, 6×6 matrix of partial derivatives, “linear transformation”

Dividing both sides by the differential time element

$\Rightarrow \dot{Y} = J(X) \dot{X}$

Jacobians are time-varying linear transformations



4. Jacobians

□ In robotics, relate joint velocities to Cartesian velocities of the tip of the arm

$${}^0\mathbf{v} = \begin{bmatrix} {}^0\mathbf{v} \\ {}^0\boldsymbol{\omega} \end{bmatrix} = {}^0J(\Theta)\dot{\Theta}$$

Θ is the vector of joint angles of the manipulator, and $\mathbf{v}$ is a vector of Cartesian velocities.

- The general case of a six-jointed robot, the Jacobian is 6×6 , $\dot{\Theta}$ is 6×1 , and ${}^0\mathbf{v}$ is 6×1 .
- This 6×1 Cartesian velocity vector is the 3×1 linear velocity vector and the 3×1 rotational velocity vector stacked together.

□ Changing a Jacobian's frame of reference (spatial motion):

Given a Jacobian written in frame $\{B\}$

$${}^B\mathbf{v} = \begin{bmatrix} {}^B\mathbf{v} \\ {}^B\boldsymbol{\omega} \end{bmatrix} = {}^BJ(\Theta)\dot{\Theta}$$

Expression for the Jacobian in another frame, $\{A\}$

$${}^A\mathbf{v} = \begin{bmatrix} {}^A\mathbf{v} \\ {}^A\boldsymbol{\omega} \end{bmatrix} = {}^AJ(\Theta)\dot{\Theta} = \begin{bmatrix} {}^A{}_B R & 0 \\ 0 & {}^A{}_B R \end{bmatrix} \begin{bmatrix} {}^B\mathbf{v} \\ {}^B\boldsymbol{\omega} \end{bmatrix}$$

Changing the frame of reference of a Jacobian is accomplished by means of this relationship

$$\Rightarrow {}^AJ(\Theta) = \begin{bmatrix} {}^A{}_B R & 0 \\ 0 & {}^A{}_B R \end{bmatrix} {}^BJ(\Theta)$$

4. Jacobians

□ Invertibility of Jacobians

$$\dot{\Theta} = J^{-1}(\Theta)v \longrightarrow \text{Calculate joint rates from given Cartesian velocities}$$

- **Singularities of the mechanism:** when the Jacobian J is NOT invertible

- **Workspace-boundary singularities**

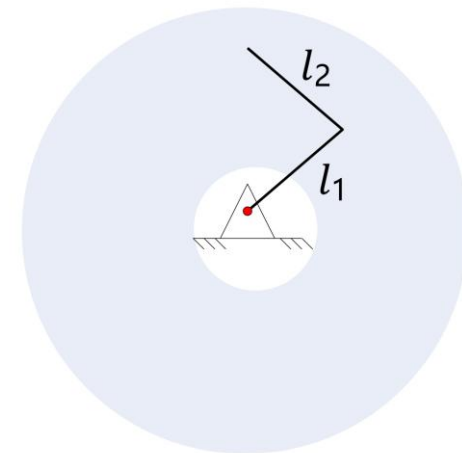
Ex: when the manipulator is fully stretch out or folded back on itself

- **Workspace-interior singularities**

Ex: occur away from the workspace boundary; generally caused by a lining up of two or more joint axes.

- **A manipulator is in a singular configuration**

- Lost one or more degrees of freedom (as viewed from Cartesian space): there is some direction (or subspace) in Cartesian space along which it is impossible to move the hand of the robot, no matter what joint rates are selected





4. Jacobians – Example: A RR Manipulator

□ **Ex:** Given a two-link manipulator with rotational joints, calculate the velocity of the tip of the arm as a function of joint velocities. Give the answer in two forms—in terms of frame {3}, and also in terms of frame {0}.

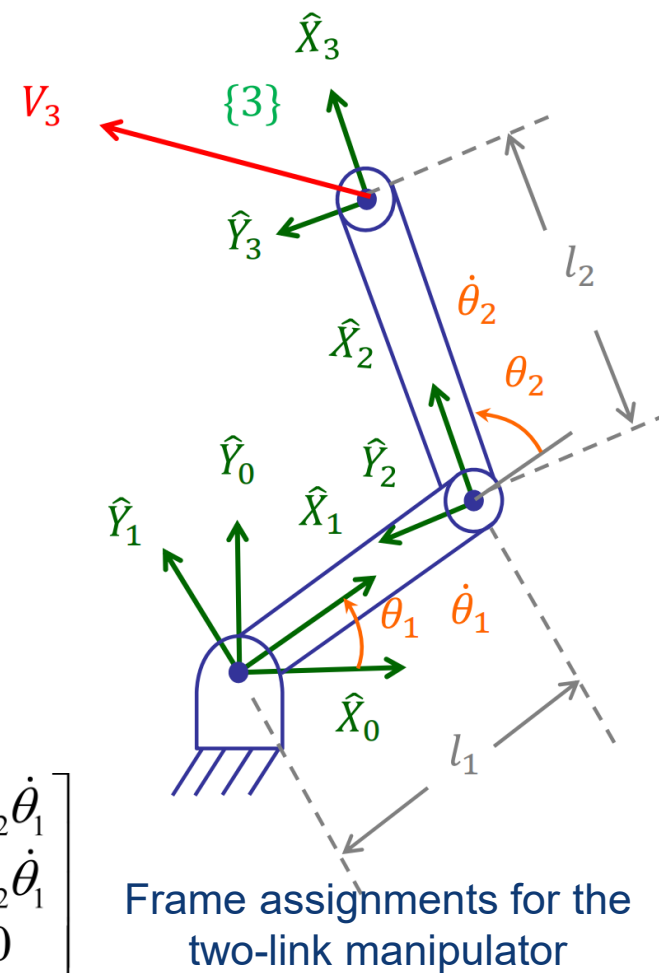
● **Method 1:** Velocity “propagation” from link to link

$${}^0_1T = \begin{bmatrix} c_1 & -s_1 & 0 & 0 \\ s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^1_2T = \begin{bmatrix} c_2 & -s_2 & 0 & l_1 \\ s_2 & c_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^2_3T = \begin{bmatrix} 1 & 0 & 0 & l_2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

● Link “propagation”

$${}^1\omega_1 = {}^1R^0 \cancel{{}^0\omega_0} + \dot{\theta}_1 {}^1\hat{Z}_1 = \dot{\theta}_1 {}^1\hat{Z}_1 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \quad {}^1v_1 = {}^1R(\cancel{{}^0v_0} + \cancel{{}^0\omega_0} \times \cancel{{}^0P_1}) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$${}^2\omega_2 = {}^2R^1\omega_1 + \dot{\theta}_2 {}^2\hat{Z}_2 = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix} \quad {}^2v_2 = {}^2R(\cancel{{}^1v_1} + {}^1\omega_1 \times {}^1P_2) = \begin{bmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ l_1\dot{\theta}_1 \\ 0 \end{bmatrix} = \begin{bmatrix} l_1s_2\dot{\theta}_1 \\ l_1c_2\dot{\theta}_1 \\ 0 \end{bmatrix}$$





4. Jacobians – Example: A RR Manipulator

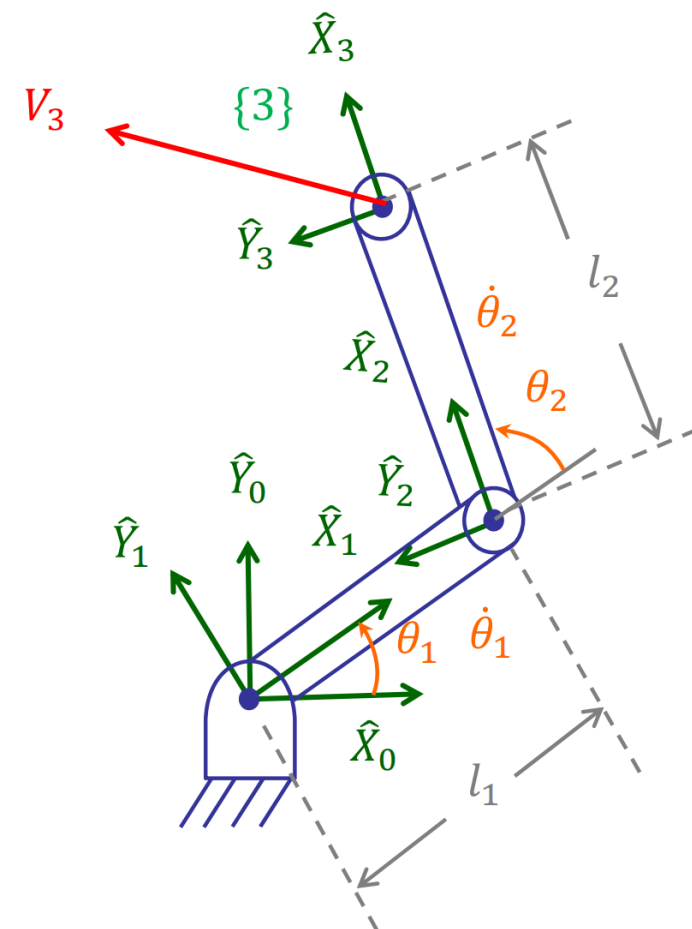
$${}^3\omega_3 = {}^2\omega_2$$

$${}^3v_3 = {}^3R({}^2v_2 + {}^2\omega_2 \times {}^2P_3) = I \left(\begin{bmatrix} l_1 s_2 \dot{\theta}_1 \\ l_1 c_2 \dot{\theta}_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix} \times \begin{bmatrix} l_2 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} l_1 s_2 \dot{\theta}_1 \\ l_1 c_2 \dot{\theta}_1 + l_2 (\dot{\theta}_1 + \dot{\theta}_2) \\ 0 \end{bmatrix}$$

To find these velocities with respect to the non-moving base frame, we rotate them with the rotation matrix 0_3R

$${}^0_3R = {}^0_1R \quad {}^1_2R \quad {}^2_3R = \begin{bmatrix} c_{12} & -s_{12} & 0 \\ s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\underline{{}^0v_3} = \underline{{}^0_3R} {}^3v_3 = \begin{bmatrix} -l_1 s_1 \dot{\theta}_1 - l_2 s_{12} (\dot{\theta}_1 + \dot{\theta}_2) \\ l_1 c_1 \dot{\theta}_1 + l_2 s_{12} (\dot{\theta}_1 + \dot{\theta}_2) \\ 0 \end{bmatrix}$$



Frame assignments for the two-link manipulator

4. Jacobians – Example: A RR Manipulator

- **Method 2:** Direct differentiation

$${}^0 \begin{bmatrix} p_x \\ p_y \\ \theta \end{bmatrix} = {}^0 \begin{bmatrix} l_1 c_1 + l_2 c_{12} \\ l_1 s_1 + l_2 s_{12} \\ \theta_1 + \theta_2 \end{bmatrix}$$

Diff.

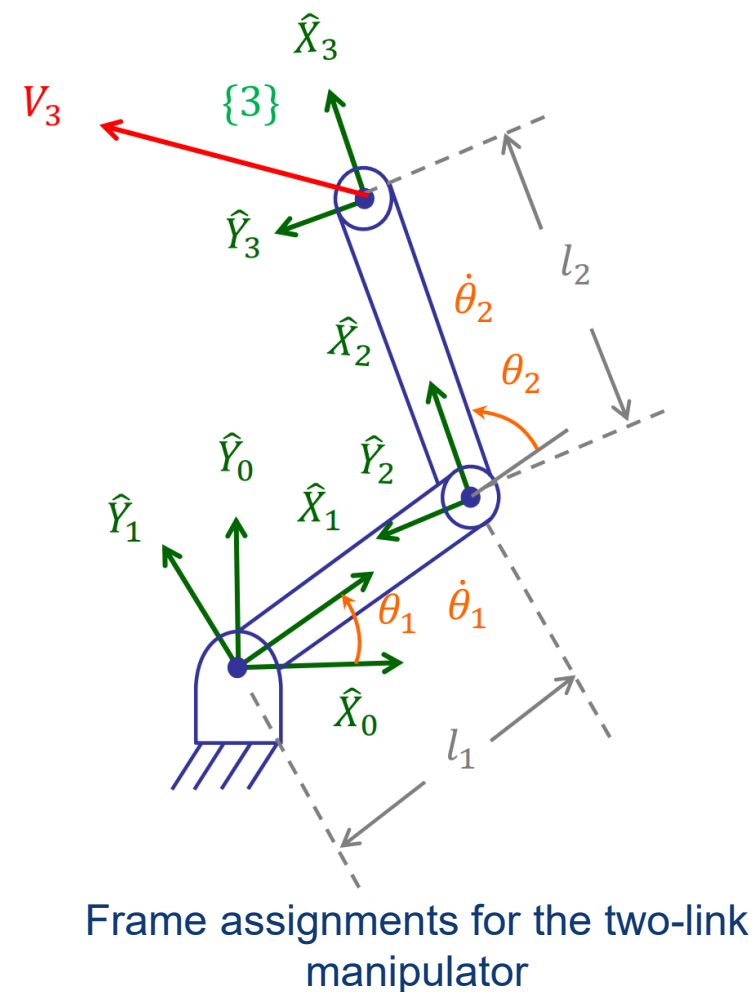
$${}^0 \begin{bmatrix} v_x \\ v_y \\ \omega \end{bmatrix} = {}^0 \begin{bmatrix} -l_1 s_1 \dot{\theta}_1 - l_2 s_{12} (\dot{\theta}_1 + \dot{\theta}_2) \\ l_1 c_1 \dot{\theta}_1 + l_2 s_{12} (\dot{\theta}_1 + \dot{\theta}_2) \\ \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix}$$

$$= {}^0 \begin{bmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ l_1 c_1 + l_2 c_{12} & l_2 c_{12} \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}$$

angular velocity of the end-effector

$$\dot{X} = {}^0 J(\Theta) \dot{\Theta}$$

The Jacobian is ${}^0 J(\Theta) = \begin{bmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ l_1 c_1 + l_2 c_{12} & l_2 c_{12} \end{bmatrix}$



4. Jacobians – Example: A RR Manipulator

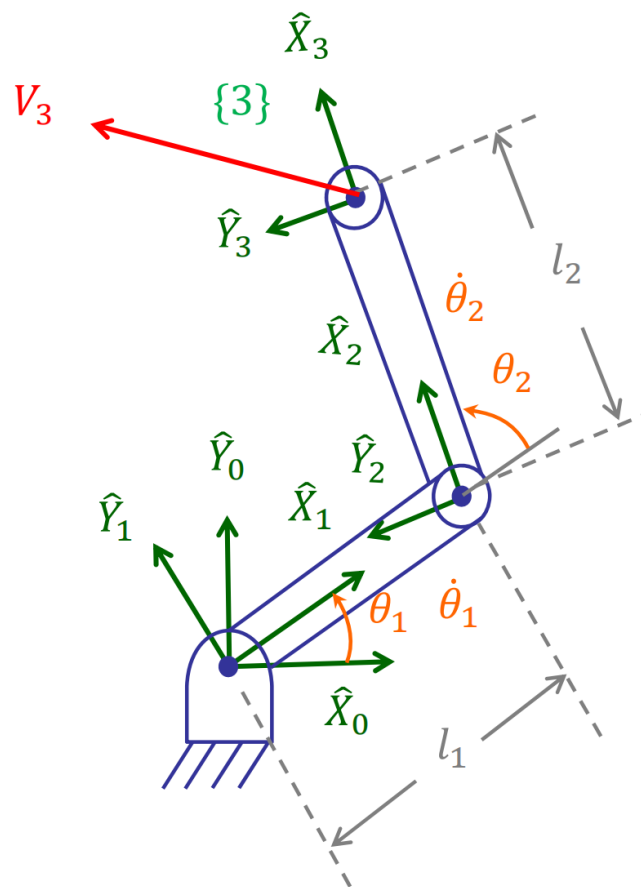
- Singularity

$${}^3\mathbf{v} = \begin{bmatrix} v_x \\ v_y \\ \omega \end{bmatrix} = \begin{bmatrix} l_1 s_2 & 0 \\ l_1 c_2 + l_2 & l_2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix} \\ = {}^3J(\Theta)\dot{\Theta}$$

Singularity configuration: $\det \begin{vmatrix} l_1 s_2 & 0 \\ l_1 c_2 + l_2 & l_2 \end{vmatrix} = l_1 l_2 s_2 = 0$

$\Rightarrow \theta_2 = 0 \text{ or } 180$

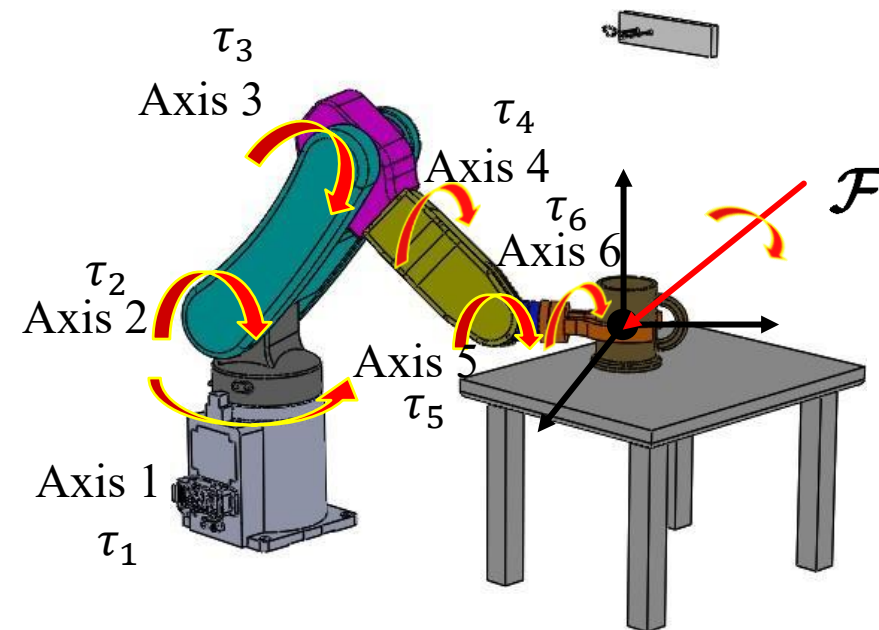
Workspace-boundary singularities



Frame assignments for the two-link manipulator

5. Static Forces in Manipulators

- The chainlike nature of a manipulator leads us quite naturally to consider how forces and moments “propagate” from one link to the next.
- Typically, the robot is pushing on something in the environment with the end-effector or is perhaps supporting a load at the hand. Wish to solve for the joint torques that must be acting to keep the system in static equilibrium.
- When considering static forces
 - Lock all the joints so that manipulator becomes a structure
 - Write force-moment relationship for each link
 - Compute static torque (ignore gravity) of joint axes



The static forces and torques considering at the joints are those caused by a **static force or torque (or both) acting on the last link.**

5. Static Forces in Manipulators

□ Static force analysis for the link

f_i : force exerted on link i by link $i - 1$

n_i : torque exerted on link i by link $i - 1$

For link $\{i\}$:

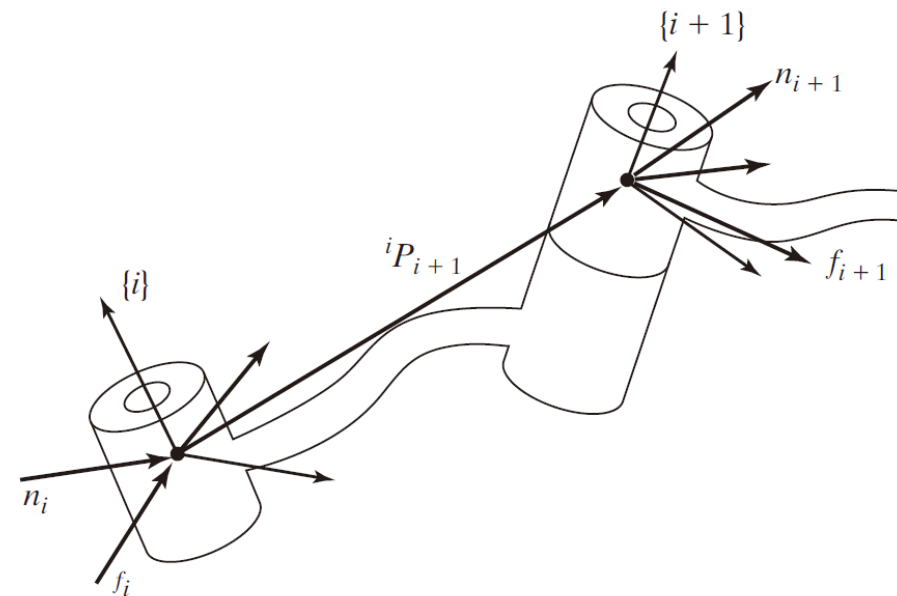
Summing the forces and setting them equal to zero,

$${}^i f_i - {}^i f_{i+1} = 0$$

Summing torques about the origin of frame $\{i\}$,

$${}^i n_i - {}^i n_{i+1} - {}^i P_{i+1} \times {}^i f_{i+1} = 0$$

Start with a description of the **force and moment applied by the robot end effector**, then calculate the force and moment applied by each link, working from the last link down to the base (link 0).



Static force-moment balance for a single link



$${}^i f_i = {}^i f_{i+1},$$

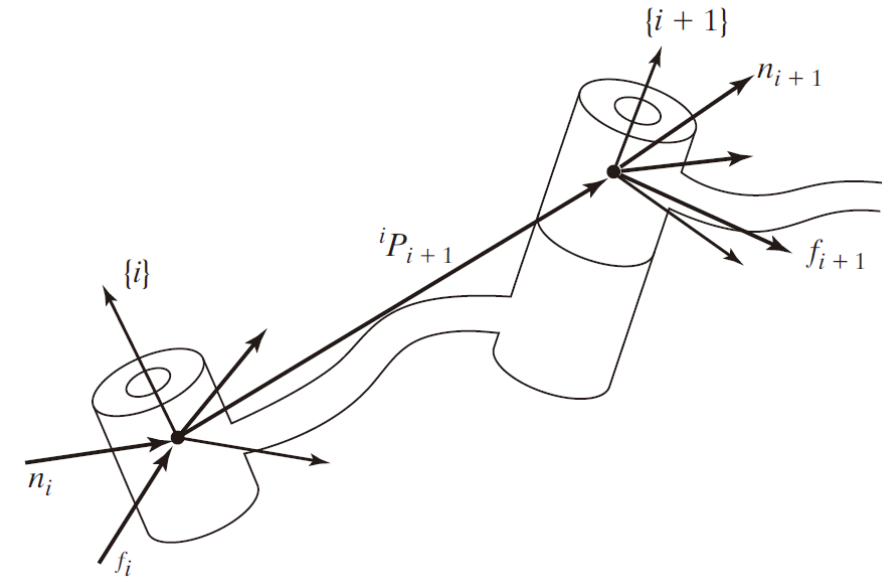
$${}^i n_i = {}^i n_{i+1} + {}^i P_{i+1} \times {}^i f_{i+1}$$

5. Static Forces in Manipulators

- Transform with the rotation matrix describing frame $\{i + 1\}$ relative to frame $\{i\}$, static force “propagation” from link to link

$$\begin{aligned} {}^i f_i &= {}^i f_{i+1} \\ &\quad \downarrow {}^{i+1}R \\ {}^i f_i &= {}^{i+1}R {}^{i+1} f_{i+1} \end{aligned}$$

$$\begin{aligned} {}^i n_i &= {}^i n_{i+1} + {}^i P_{i+1} \times {}^i f_{i+1} \\ &\quad \downarrow {}^{i+1}R \\ {}^i n_i &= {}^{i+1}R {}^{i+1} n_{i+1} + {}^i P_{i+1} \times {}^i f_i \end{aligned}$$



Static force-moment balance for a single link

- All components of the force and moment vectors are resisted by the structure of the mechanism itself, except for the torque about the joint axis.
- To find the joint torque required to maintain the static equilibrium, the dot product of the joint-axis vector with the moment vector acting on the link is computed:
 - Revolute joint, moment vector $\tau_i = {}^i n_i^T {}^i \hat{Z}_i$
 - Prismatic joint, joint actuator force $\tau_i = {}^i f_i^T {}^i \hat{Z}_i$

5. Static Forces in Manipulators: Example - A RR Manipulator

□ Force “propagation” from link to link

Starting from the last link and going toward the base of the robot

$${}^2f_2 = {}^2_3R {}^3f_3 = I {}^3F = \begin{bmatrix} f_x \\ f_y \\ 0 \end{bmatrix} \quad {}^2n_2 = {}^2_3R {}^3n_3 + {}^2P_3 \times {}^2f_2 = \begin{bmatrix} 0 \\ 0 \\ l_2 f_y \end{bmatrix}$$

$${}^1f_1 = {}^1_2R {}^2f_2 = \begin{bmatrix} c_2 & -s_2 & 0 \\ s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f_x \\ f_y \\ 0 \end{bmatrix} = \begin{bmatrix} c_2 f_x - s_2 f_y \\ s_2 f_x + c_2 f_y \\ 0 \end{bmatrix}$$

$${}^1n_1 = {}^1_2R {}^2n_2 + {}^1P_2 \times {}^1f_1 = \begin{bmatrix} 0 \\ 0 \\ l_1 s_2 f_x + l_1 c_2 f_y + l_2 f_y \end{bmatrix}$$

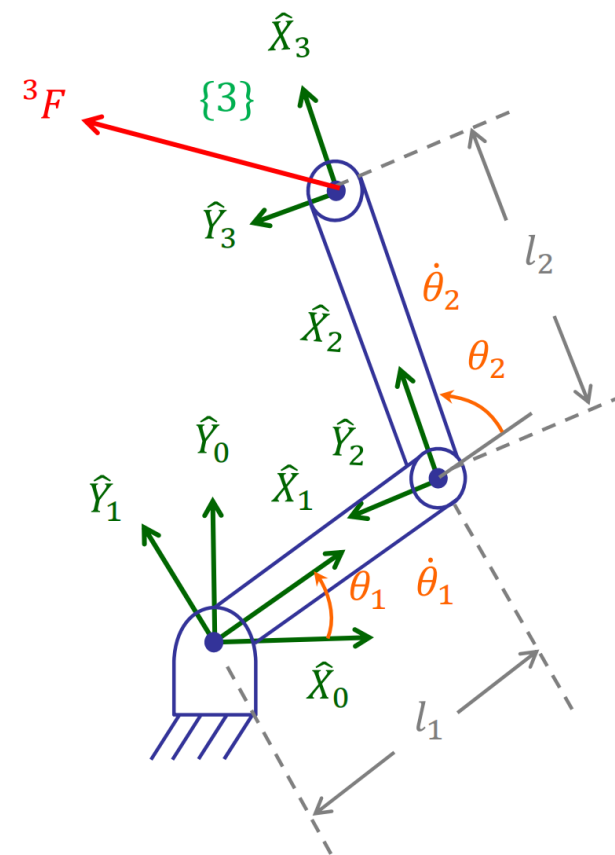
□ Therefore,

$$\tau_1 = {}^1n_1^T {}^1\hat{Z}_1 = l_1 s_2 f_x + (l_2 + l_1 c_2) f_y$$

$$\tau_2 = {}^2n_2^T {}^2\hat{Z}_2 = l_2 f_y$$



$$\begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} l_1 s_2 & l_2 + l_1 c_2 \\ 0 & l_2 \end{bmatrix} \begin{bmatrix} f_x \\ f_y \end{bmatrix}$$



6. Jacobian in the Force Domain

- The principle of virtual work: equate the work done in Cartesian terms with the work done in joint-space terms **given infinitesimal displacement**

$$\mathcal{F} \cdot \delta \chi = \tau \cdot \delta \Theta$$

where, $\mathcal{F}$ is a 6×1 Cartesian force-moment vector acting at the end-effector, $\delta \chi$ is a 6×1 infinitesimal Cartesian displacement of the end-effector, τ is a 6×1 vector of torques at the joints, $\delta \Theta$ is a 6×1 vector of infinitesimal joint displacements.

Also be written as $\mathcal{F}^T \delta \chi = \tau^T \delta \Theta$

The definition of the Jacobian is $\delta \chi = J \delta \Theta$

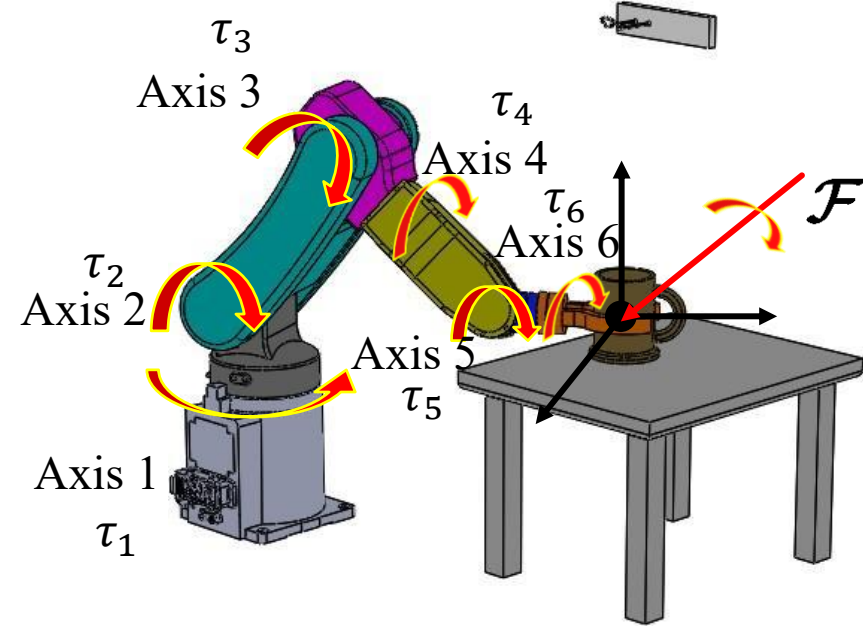
Write it as $\mathcal{F}^T J \delta \Theta = \tau^T \delta \Theta$

↓ Must hold for all $\delta \Theta$

$$\mathcal{F}^T J = \tau^T$$

Transposing both sides yields this $\tau = J^T \mathcal{F}$ → $\tau = {}^0 J^T {}^0 \mathcal{F}$

The Jacobian transpose maps Cartesian forces acting at the hand into equivalent joint torques



When the Jacobian loses full rank, there are certain directions in which the end effector cannot exert static forces even if desired:

- If the Jacobian is singular, $\mathcal{F}$ could be increased or decreased in certain directions without effect on the value calculated for τ .
- Near singular configurations, mechanical advantage tends toward infinity, such that, with small joint torques, large forces could be generated at the end-effector

Jacobian is written with respect to frame {0}



7. Cartesian Transformation of Velocities and Static Forces

□ 6×1 representations of **general velocity** and **general force representations**

$$\mathbf{v} = \begin{bmatrix} \mathbf{v} \\ \omega \end{bmatrix}$$

$$\mathcal{F} = \begin{bmatrix} F \\ N \end{bmatrix}$$

□ Frame transformation

$$\begin{aligned} {}^{i+1}\omega_{i+1} &= {}^{i+1}_i R {}^i\omega_i + \dot{\theta}_{i+1} {}^{i+1}\hat{Z}_{i+1} \\ {}^{i+1}\mathbf{v}_{i+1} &= {}^{i+1}_i R ({}^i\mathbf{v}_i + {}^i\omega_i \times {}^i\mathbf{P}_{i+1}) \end{aligned}$$

Propagation of velocities and forces from link to link

↓ $i = B, i+1 = A, \dot{\theta}_{i+1} = 0$ ← Two frames are rigidly connected

$$\begin{bmatrix} {}^A\mathbf{v}_A \\ {}^A\omega_A \end{bmatrix} = \begin{bmatrix} {}^A_B R & {}^A\mathbf{P}_{B\text{ORG}} \times {}^A_B R \\ 0 & {}^A_B R \end{bmatrix} \begin{bmatrix} {}^B\mathbf{v}_B \\ {}^B\omega_B \end{bmatrix}$$

$${}^A\mathbf{v}_A = {}^A T_v {}^B\mathbf{v}_B \quad \mathbf{P} \times = \begin{bmatrix} 0 & -p_z & p_y \\ p_z & 0 & -p_x \\ -p_y & p_x & 0 \end{bmatrix}$$

Cross product is understood to be the matrix operator

Write the 6×6 matrix that transforms general force vectors written in terms of $\{B\}$ into their description in frame $\{A\}$

$$\begin{bmatrix} {}^A F_A \\ {}^A N_A \end{bmatrix} = \begin{bmatrix} {}^A_B R & 0 \\ {}^A\mathbf{P}_{B\text{ORG}} \times {}^A_B R & {}^A_B R \end{bmatrix} \begin{bmatrix} {}^B F_B \\ {}^B N_B \end{bmatrix}$$

Force-moment transformation.

$${}^A\mathcal{F}_A = \boxed{{}^A T_f} {}^B\mathcal{F}_B$$

Similarly, ${}^i f_i = {}^i f_{i+1}$,
from ${}^i n_i = {}^i n_{i+1} + {}^i \mathbf{P}_{i+1} \times {}^i f_{i+1}$

Mappings of velocities from frame to frame depend on ${}^A T_B$

$$\begin{bmatrix} {}^B\mathbf{v}_B \\ {}^B\omega_B \end{bmatrix} = \begin{bmatrix} {}^B_A R & -{}^B_A R {}^A\mathbf{P}_{B\text{ORG}} \times \\ 0 & {}^B_A R \end{bmatrix} \begin{bmatrix} {}^A\mathbf{v}_A \\ {}^A\omega_A \end{bmatrix}$$

$${}^B\mathbf{v}_B = {}^B T_v {}^A\mathbf{v}_A$$

Inverse



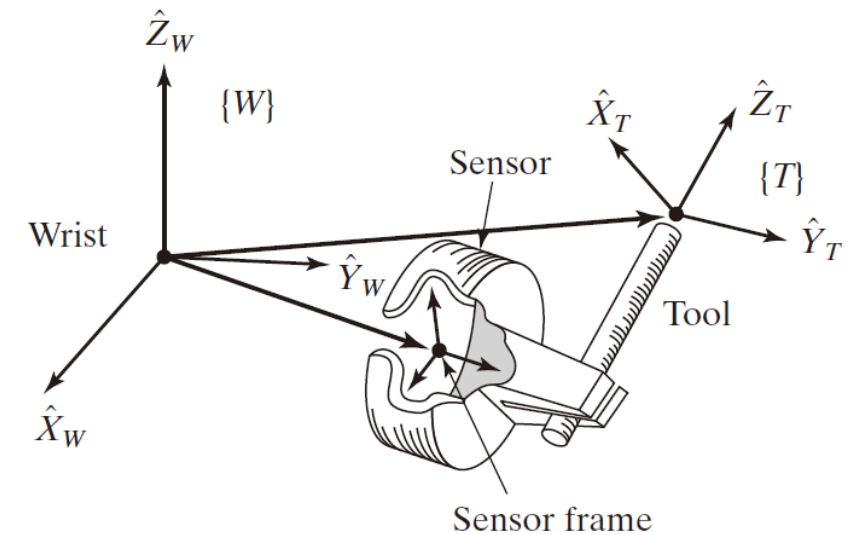
7. Cartesian Transformation of Velocities and Static Forces

■ **Ex:** Figure shows an end-effector holding a tool. Located at the point where the end effector attaches to the manipulator is a force-sensing wrist. This is a device that can measure the forces and torques applied to it. Consider the output of this sensor to be a 6×1 vector, ${}^S\mathcal{F}$, composed of three forces and three torques expressed in the sensor frame, $\{S\}$. Our real interest is in knowing the forces and torques applied at the tip of the tool, ${}^T\mathcal{F}$. Find the 6×6 transformation that transforms the force-moment vector from $\{S\}$ to the tool frame, $\{T\}$. The transform relating $\{T\}$ to $\{S\}$, S_TT , is known. (Note that $\{S\}$ here is the sensor frame, not the station frame.)

First, from S_TT , we calculate the inverse, ${}^T_S T$, which is composed of ${}^T_S R$ and ${}^T P_{S\text{ORG}}$.

Then, we apply ${}^A\mathcal{F}_A = {}^A_B T_f {}^B\mathcal{F}_B$ and obtain

$${}^T\mathcal{F}_T = {}^T_S T_f {}^S\mathcal{F}_S \quad \text{where, } {}^T_S T_f = \begin{bmatrix} {}^T_S R & 0 \\ {}^T P_{S\text{ORG}} \times {}^T_S R & {}^T_S R \end{bmatrix}$$



Frames of interest with a force sensor



Summary

- ❑ The representation methods for linear and angular velocities of rigid bodies.
- ❑ The analysis methods for velocity “Propagation” from link to link, and the definition and significance of the Jacobian matrix.
- ❑ The derivation of Jacobians in the force domain.
- ❑ The mapping between velocity and static forces through cartesian transformation.

Jacobians Velocities and Static Forces

- Time-varying Position and Orientation
- Linear and Rotational Velocity of Rigid Bodies
- Velocity “Propagation” from Link to Link
- Jacobians
- Static Forces in Manipulators
- Jacobian in the Force Domain
- Cartesian transformations between velocity and static forces



Q&A